

Qualification Task Part 1

Over the last few years Python has emerged to be the most popular language in the data science field. Therefore, please first familiarize yourself with the following content. You should follow the videos very closely and program along with them.

At our next meeting please show us a GitHub repository where you present your `.py` files with the contents and comments about their functionalities in code as well as any little modifications of your own choice. Please document everything and present reproducible results.

Choose a Python distribution and setup an environment:

- <https://www.python.org/>
- <https://docs.conda.io/projects/miniconda/en/latest/index.html>
- <https://www.anaconda.com/>

Choose one Integrated Development Environment:

- VS-Code: <https://code.visualstudio.com/>
- PyCharm
 - https://www.youtube.com/watch?app=desktop&v=hc50ALh_x5g
 - <https://www.jetbrains.com/pycharm/>
- Spyder:
 - <https://www.youtube.com/playlist?app=desktop&list=PLPonohdiDqg9epCIEcXoAPUiK0pN5eRoc&cbrd=1>

Programming Style guides: good to know about

- <https://peps.python.org/pep-0008/#introduction>
- https://visualgit.readthedocs.io/en/latest/pages/naming_convention.html
- <https://realpython.com/python-pep8/>

Basic Python

- <https://www.youtube.com/watch?v=YYXdXT2l-Gg&list=PL-osiE80TeTskrapNbzxhwoFUiLCjGgY7>
- https://www.youtube.com/watch?v=nLRL_NcnK-4

Git

- <https://www.youtube.com/watch?app=desktop&v=RGQj5yH7evk>

Testing

- <https://www.youtube.com/watch?v=6tNS--WetLI>

- <https://www.youtube.com/watch?v=cHYq1MRoyl0>
- <https://docs.python.org/3/library/unittest.html>
- <https://www.youtube.com/watch?v=YbpKMIUjvK8>

Numpy

- <https://www.youtube.com/watch?app=desktop&v=QUT1VHiLmml>
- <https://www.youtube.com/watch?app=desktop&v=DcfYgePyedM>
- <https://www.youtube.com/watch?app=desktop&v=ZB7BZMhfPgk>
- <https://numpy.org/>

Matplotlib and Seaborn

- <https://www.youtube.com/watch?app=desktop&v=3Xc3CA655Y4&pp=ygUTbWF0cGxvdGxpYiB0dXRvcmlhbmA%3D%3D>
- <https://matplotlib.org/>
- <https://www.youtube.com/watch?v=6GUZXDef2U0>
- <https://seaborn.pydata.org/>

Qualification Task Part 2

In the following task we use only plain Python, NumPy, and the introduced graphics libraries except otherwise stated.

Please develop a Python class of a Robust Data Science student affine to numerical programming.

The class itself is a descendant of a TU Darmstadt student class with properties such as name, age, date of student registration at the university, study program, registration number, 5 accomplished courses, 1 favorite course.

We consider name, age, date of registration, study program and registration to be private attributes, as such getter methods should be supplied to return the attribute values. The course list and the favorite course should have getter and setter methods.

The class should also have associated unit tests to check all functionalities.

The descendant Robust Data Science student class should provide additionally three methods which cope with the following problems:

Solve an integral problem: Take an input argument x representing a number range such as $x \in [0, 12]$. Furthermore, it should take another interval for statistical calculus such as $x_{stats} \in [4, 7]$. Internally, the method should compute:

- $y = e^{-x} \cos(x)$ and plot the function.
- Compute the mean, variance and standard deviation of y values for x_{stats} .
- For x_{stats} find the threshold value y_m s.t. 70% of the y values are below the threshold.
- Plot dy/dx for x if an additional Boolean input argument allows it.
- Find the index of $\frac{dy}{dx} = 0$.

- The method should perform a meaningful input type check before any calculus is executed. Otherwise, the executions should be passed.
- The method should give numerical results as a nice output on the console.

As second method, the class should supply functionality to solve a linear algebra problem for inputs of a system matrix and a solution vector. As test case you can consider the following examination of a resistor network where the voltages $V_i, i \in [1,4]$ are unknown:

$$\begin{aligned} 3V_1 + 2V_2 + 3V_3 + 10V_4 &= 4 \\ 2V_1 - 2V_2 + 5V_3 + 8V_4 &= 1 \\ 3V_1 + 3V_2 + 4V_3 + 9V_4 &= 3 \\ 3V_1 + 4V_2 - 3V_3 - 7V_4 &= 2. \end{aligned}$$

The method should perform a meaningful input type check before any calculus is executed. Otherwise, the executions should be passed. As before a console output should summarize the results of your solution vector.

Finally, add a method which solves a multivariate least-squares linear regression problem for inputs of the regressor matrix and the response vector.

The function should take as inputs the response variable y and the regressor matrix X . As restriction, the number of observations should always excess the number of variables.

We aim for outputs of:

- The coefficient values for the parameter vector beta.
- Their t-statistics and p-values to assess their statistical significance.
For this task the SciPy library is allowed to be used.
- An optional console output of the numerical results depending on a Boolean input.

The method should perform a meaningful input type check before any calculus is executed. Otherwise, the executions should be passed.

A script should demonstrate the applicability of the two classes. Also, both classes should be well documented.

The 2nd qualification task should also be presented in its own GitHub repository.